

Shukai Huang

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Software engineer completing a Master of Information Technology at the University of Auckland, with four years of backend development experience in live-service systems. Experienced in **backend development**, **production delivery**, **distributed systems design** and **server-side optimization**. Recent projects focus on **full-stack web applications**, **AI-assisted workflows** and **cloud/serverless delivery on AWS**.

TECHNICAL SKILLS

Programming	TypeScript, Python, C#, C++, Java, Erlang
Web & APIs	HTML, CSS, React, RESTful APIs, Express/Node.js, FastAPI, ASP.Net
SQL & Databases	SQL, MySQL, PostgreSQL, MongoDB
Cloud & Delivery	Azure, AWS, Docker, GitHub Actions, Git, Linux, Nginx
AI & LLMs	LangGraph, Tool Calling, Agent Workflows, Prompt Engineering, RAG, MCP

PROFESSIONAL EXPERIENCE

4399 Information Technology, Guangzhou, China Software Engineer September 2023 — April 2024

Tech Stack: TypeScript, Erlang, Cocos2d, Protobuf, MySQL

- **Systems Integration:** Built cross-server event workflows between regional game servers and central services.
- **Backend Delivery:** Delivered features across multiple projects, versions and release branches.
- **Production Support:** Traced service logs, checked database records and applied hotfixes or correction scripts.
- **Cross-team Collaboration:** Worked with product, QA and operations teams on regional requirements and weekly releases.

Leniu Technology, Guangzhou, China Backend Developer June 2020 — May 2023

Tech Stack: Python, Erlang, Lua, MySQL, Jenkins, SVN

- **Full SDLC:** Contributed from early development to beta testing and live operation.
- **Feature Ownership:** Led backend delivery from requirements and design through testing and release.
- **Automation:** Optimised a Python configuration tool, reducing manual scripting by 50% and generated code size by 30%.
- **System Reliability:** Improved caching, scheduled-event logic, test validation and production troubleshooting.

PROJECTS

PetPal - AI-powered pet care communication platform Backend / Full-Stack Developer July 2026 — Present

Tech Stack: Python, FastAPI, LangGraph, PostgreSQL, Docker, AWS Lambda, API Gateway, Aurora, GitHub Actions

- **LangGraph Agents:** Engineered parent/staff workflows with scoped tool calls, care-evidence validation and human handoff.
- **Reliable AI Actions:** Enforced authorization and idempotent FastAPI/PostgreSQL writes to prevent duplicate AI actions.
- **AWS Serverless:** Deployed ECR images to API/worker Lambdas with API Gateway and IAM-authenticated Aurora PostgreSQL.
- **CI/CD:** Built GitHub Actions/OIDC pipelines with immutable image digests, Alembic migrations and release health checks.
- **Web Delivery:** Published frontend/media through S3/CloudFront with automated asset sync and CDN cache invalidation.
- **Background Jobs:** Scheduled Lambda via EventBridge for task deadlines, stay expiry and escalation follow-ups.

UFoundIt - Campus exploration and photo check-in platform Backend / Full-Stack Developer March 2026 — May 2026

Tech Stack: React, TypeScript, MongoDB, Azure, Docker, GitHub Actions, Nginx, OpenAI API

github.com/Treekay/UFoundIt

- **Full-stack Delivery:** Built React flows and APIs for challenges, check-ins, reviews and leaderboards.
- **Map-based UX:** Added Leaflet/OpenStreetMap selection for check-in coordinates and submissions.
- **AI Validation:** Designed image review with match checks, duplicate detection and confidence thresholds.
- **Cloud Deployment:** Deployed on Azure with Docker Compose, Nginx, HTTPS and GitHub Actions.

EDUCATION

University of Auckland – Master of Information Technology July 2025 — November 2026 (Expected)

Sun Yat-sen University – Bachelor of Software Engineering September 2016 — June 2020

TRAINING & CREDENTIALS

AWS Academy Graduate – Cloud Architecting – Training Badge November 2025

AWS Academy Graduate – Cloud Foundations – Training Badge October 2025

ADDITIONAL EXPERIENCE

Woolworths, Auckland Retail Assistant February 2026 – Present

Local Teamwork: Support time-sensitive store operations, stock replenishment and customer-facing coordination.

INTERESTS

Guitar; Badminton; Hiking with my Corgi.